

2022 UT Austin Invitational
October 29, 2022

Dynamic Planet Division B Answer Key



Important Notice

- **YOU SHOULD NOT HAVE THIS! IF YOU DO, DO NOT OPEN AND NOTIFY THE EVENT SUPERVISORS IMMEDIATELY!**
- Failure to disclose the event supervisors on obtaining this will result in an immediate disqualification.

Directions

- All answers must be indicated on your answer document; test booklets will NOT be graded.
- Point values are indicated for all questions and tiebreakers are indicated in test booklets.

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(any questions after the conclusion of this event may be emailed to lalithroop@gmail.com and crystal@atxscioly.org)

Page:	Points
2	26
3	28
4	23
5	12
Total	90

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1. C (+1)
2. B (+1)
3. B (+1)
4. B (+2 for correct answer)
5. +1 for showing work: $10,000/9,000$
+1 for answer: 1.11
6. C, D (+2 for having this exact answer)
7. C (+1)
8. C (+1)
9. C (+1)
10. Rocks from canyon sides (+1) fell after freeze-thaw wedging (+1) from glaciation (+1)
11. +0.5 for each correct answer
 - a. Decrease
 - b. Increase
 - c. Increase
 - d. Decrease
 - e. Decrease
 - f. Decrease
12.
 - a. $2.094 \times 10^9 \text{ ft}^3$ (+1 for correct answer)
 - b. +1 for showing surface area calculation: Surface Area of lake = $\pi r^2 = \pi (1000)^2 = 3.142 \times 10^6 \text{ ft}^2$
+1 for correct answer: $15/12 \times \text{SA} = \mathbf{3.927 \times 10^6 \text{ ft}^3/\text{yr}}$
 - c. +2 for correct answer: $15 \times 3 \times 30 = 1350 \text{ ft}^3/\text{s} \times 3600 \times 24 \times 365 = \mathbf{4.257 \times 10^{10} \text{ ft}^3/\text{yr}}$
 - d. Fall, more outflow than inflow (+1)
 - e. +2 for correct answer: $(2.094 \times 10^9) / (4.0 \times 10^{10}) \times 365 = \mathbf{19.108 \text{ days}}$

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13. 70.9 N 4.3 E (+1)
14. Disappearing stream (+1)
15. A (+1)
16. D (+1)
17. Carbonates are better eroded by acidic water (+1) which is common at the surface due to acid rain/pollution (+1)
- 18.

- a. Porosity due to packing structure (+1). There is 0.39 void space for every 1 unit of total volume (+1).
 - b. Greater (+1) secondary porosity is from additional fracturing (+1)
 - c. Area of unit sphere is $\frac{4}{3} \pi r^3$ while cube is $8r^3$ so take ratio. $1 - (\frac{4}{3} \pi)/8 = 0.4764$ (+1 for answer, +2 for acceptable work)
19. B (+1)
20. A (+1)
21. False (+1). Pore water pressure is lower than atmospheric pressure in the vadose zone (unsaturated zone) due to capillary forces attracting water to mineral surfaces (+1).
22. Thicker (+1). In materials with smaller pores, like clay and silt, there's more surface area and greater surface attraction forces (+1) .
23. C (+1)
24. A (+1)
25. C (+1) - Artesian Well (+1)
26. B (+1) - Potentiometric Surface (+1)
27. Soggy wetlands, springs, points, and streams (+1). The recharge rate is high relative to transmissivity here, causing the water table to be just under the surface or intersecting the surface. (+1)
28. Effluent streams gain water from the saturated zone, while influent streams lose water to the saturated zone. When the water table is high, the saturated zone is higher, so streams are more likely to gain water and vice versa. (+1)

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29. B (+1)
30. C (+1)
31. A (+1)
32. D (+1)
33. B (+1)
34. Active Layer (+1)
35. Low average temperature, shady slopes (north-facing ones in the Northern Hemisphere), little vegetation to shelter the ground surface from winds, little snow cover to insulate the ground, and/or drier soil (+1 for each; max at 3)
36. B, C (+1 for this exact answer)
37. E (+1)
38. C (+1)
39. A (+1)
40. B (+1)

41. A (+1)

42. B (+1)

43.

- a. Goes from increasing to decreasing (+1)
- b. Low, pretty much 0 (+2)
- c. Chemical weathering (+1); equations show mechanical weathering will always decrease (+1)
- d. More water can get in (+2)

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44. B (+1)

45.

- a. B (+1)
- b. Higher, greater, any version of that (+1)
- c. Evaporation (+1)
- d. Willows are plants that use a large amount of groundwater, so during the growing season, the water table may be locally depressed below the lake level (+1). While outflow will occur at the margins of the lake, inflow still occurs beneath the bottom (+1).
- e. The well was contaminated by a faulty/damaged septic tank, which contains waste water (+1). The waste water likely seeped into the groundwater supply (+1).
- f. In seepage lakes that are surrounded by the water table on all sides, the stagnation zone leads to upward seepage through the lake bottom. As long as the stagnation zone is present, the lake will never lose water through the bottom (Answer just needs to include: that the lake will never lose water through the bottom OR leads to upward seepage through the lake bottom) (+1)

46. True (+1)

47. A, B, C, D (+2 for this exact answer)